Annapragada Vamsy Vrishank

+1 201-936-2248 • vamsy.vrishank@gmail.com • LinkedIn, GitHub, vamsy.org

Software/Data Engineer with 3.5+ years of experience building high-throughput data infrastructure. Currently specializing in applied AI for technology, engineering the underlying platforms, data pipelines, and software that power intelligent applications.

Experience

- **PayPal** — **Bangalore, India**
  Software Engineer II (Unified Enterprise Customer) — *Jul 2024 – Jan 2025*
  - Unified 200M+ customer profiles across Braintree, Venmo, and Hyperwallet using probabilistic identity resolution; built normalized data schemas and BigQuery golden records empowering enterprise risk and ML pipelines.
  - Engineered near-real-time event streaming pipelines (Java, GCP Pub/Sub, Dataflow) across 500K+ daily transactions; slashed end-to-end data latency from 6 hours to 10 minutes for downstream fraud monitoring and analytics.
  - Standardized production Airflow DAG architecture across 100+ pipelines; authored reusable DAG patterns and testing harnesses, reducing operational toil and accelerating engineering onboarding.
- **PayPal** — **Bangalore, India**
  Software Engineer I (Enterprise Data Lake) — *Sep 2021 – Jul 2024*
  - Architected and deployed a concurrent Java microservice to resolve connection exhaustion across 300+ concurrent Airflow pipelines, eliminating operational failures and driving org-wide adoption.
  - Re-platformed 800TB+ enterprise data lake infrastructure from on-prem Hadoop/Spark to GCP; developed and maintained 200+ production ETL/ELT pipelines with Spark and Pub/Sub in Python and Java.
  - Migrated Braintree's data warehouse to BigQuery with query-pattern-aligned partitioning; cut operational reporting costs by 60% ($4M annual savings) and delivered scalable dimensional models for analytics and research teams.
- **Indian Institute of Technology (BHU)** — **Varanasi, India**
  Research Intern — *May 2019 – Jul 2019*
  - Processed 150+ hours of high-frequency sensor and EEG time series; engineered signal filtering pipelines in Python and validated CNN architectures for multi-class signal pattern recognition.

Projects

- **StoryForge: Multi-Agent LLM Pipeline**: Built an AI application orchestrating three LLM agents with stateful context and automated evaluation (LLM-as-a-judge); served via FastAPI and deployed on Google Cloud Run.
- **Distributed Log Aggregation Pipeline**: Developed a real-time observability pipeline using Kafka for stream ingestion, PyFlink for data filtering, and Elasticsearch/Kibana for fast search indexing and visualization.
- **Execution-Aware Alpha Modeling (Capstone)**: Optimized trading schedules and decaying alpha signals using an augmented Almgren-Chriss framework; derived closed-form solutions and statistically validated performance gains over standard models.
- **C++ Derivative Pricing Engine**: Engineered a multi-asset pricing engine implementing Black-Scholes, Heston, and SABR models alongside Monte Carlo simulations and PDE solvers; optimized numerical methods using C++ and Boost.

Education

- **Stevens Institute of Technology** — **Hoboken, NJ, United States**
  Masters in Financial Engineering (CGPA: 3.934/4.0) — *Jan 2025 – May 2026*
- **National Institute of Technology (NIT), Trichy** — **Tamil Nadu, India**
  Bachelors in Metallurgical and Materials Engineering, Minor: Humanities (CGPA: 8.07/10.0) — *Jul 2017 – May 2021*

Technical Skills

- **Languages & Core Libraries**: Python (FastAPI, Pandas, NumPy, pytest), C++ (Boost, STL), Java (Spring Boot), SQL.
- **Data Engineering & Distributed Systems**: GCP (BigQuery, Pub/Sub, Dataflow, Dataproc), Kafka, Apache Spark, PyFlink, Apache Airflow, Change Data Capture (CDC), Batch & Stream Processing.
- **AI Engineering & Quantitative Modeling**: LLM Pipelines (Google ADK, Gemini API), Multi-Agent Systems, RAG, Quantitative Execution Frameworks, Algorithm Optimization.
- **Infrastructure & DevOps**: Docker, Kubernetes, CI/CD, Terraform, Linux/Unix, Server-Sent Events (SSE), RESTful APIs, Microservices Architecture.
- **Databases & Observability**: PostgreSQL, Redis, Elasticsearch, Kibana, Snowflake, Schema Modeling.